

MATLAB

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5G AND 6G LABLAB EXPERIMENTNew FolderASSESSMENT QNO 1.mEX: 2.mex 1.mEX 2 .mEX 2.mEx 2.mex 5.mex 1.mWorkspace

Figure 1

5G NR Frame Structure for Different Subcarrier Spacing

Command Window

36°CPartly sunny14:5003-08-2026

FIGURE

Labels and Annotations

Figure 1

5G NR Frame Structure for Different Subcarrier Spacing

Bar chart showing Slots per 10 ms Frame vs Subcarrier Spacing (kHz). The x-axis ranges from 15 to 120 kHz, and the y-axis ranges from 0 to 80 slots. The data points are: 15 kHz (10 slots), 30 kHz (20 slots), 60 kHz (40 slots), and 120 kHz (80 slots).

Subcarrier Spacing (kHz)	Slots per 10 ms Frame
15	10
30	20
60	40
120	80

EX1.m | ex2.m | ex3.m | EX4.m | untitled4.m | EX6.m

```
1 clc; clear; close all;
2
3 % 5G NR Frame Structure Analysis with Different Subcarrier Sp
4
5 SCS = [15 30 60 120]; % Subcarrier Spacing (kHz)
6 FrameDuration = 10; % Frame Duration (ms)
7 SlotsPerFrame = SCS/15 * 10;
8
9 fprintf('5G NR Frame Structure Analysis\n');
10 fprintf('-----\n');
11 fprintf('SCS(kHz)\tSlots/Frame\tSlot Duration(ms)\n');
12
13 for i = 1:length(SCS)
14
15     SlotDuration = FrameDuration / SlotsPerFrame(i);
16
17     fprintf('%d\t%d\t%.4f\n', ...
18         SCS(i), SlotsPerFrame(i), SlotDuration);
19
20 end
21
```

SCS(kHz)	Slots/Frame	Slot Duration(ms)
15	10	1.0000
30	20	0.5000
60	40	0.2500
120	80	0.1250